Pakistan fuel prices, with receipts

A governed dataset, a baseline-beating risk model, and a refusal-first insight agent

Problem

Pakistani petrol and diesel prices are set at fortnightly reviews and felt immediately by households and businesses. Evidence is scattered across weekly PBS reports, city-level annexes and global benchmark feeds. We built a system that answers fuel-price questions with a citation on every number, and refuses when the evidence does not support an answer.

Dataset (all rows from original publishers)

PBS weekly SPI: 41 weeks x 2 products (petrol super, high-speed diesel). PBS monthly city annex: 18 reporting locations (17 cities + national average) x 2 products. US EIA: 11 daily petroleum spot series, 84,546 observations. World Bank Pink Sheet: 6,720 monthly energy observations. Total clean layer: 91,384 rows. Every raw file is SHA-256 hashed in data/manifest.csv and every gold fact stores its source row ids and citation tag. No synthetic, interpolated or re-uploaded data.

Model: next-month Brent volatility

Baseline persistence MAE 17.80; challenger har_ridge_alpha_10 MAE 16.51. Improvement 7.27% on a chronological 80/20 split (369 train / 93 test, 2019-02-01 to 2026-10-01). R2 improved from 0.008 to 0.113. The nominal 90% interval has 89.2% empirical coverage. Errors are published by year, month and volatility regime.

Reported, not claimed

Monthly average Brent is close to a random walk: persistence MAE 6.10 USD/bbl vs challenger 5.89 USD/bbl. We report this rather than presenting a spurious price-level win.

Insight agent

Vector retrieval over 2,129 gold facts (Chroma when an index is present; in-memory TF-IDF otherwise). Answers quote retrieved facts; the agent never generates a number. A declared-scope gate, a retrieval-confidence floor and time/product/city filters make refusal a code path, not a prompt request. Out-of-scope questions ('weather in Lahore') and unsupported periods ('January 2024') return 'I don't know'. Every answer returns citation tags and row ids. English and Urdu are supported.

Automation and reproducibility

Prefect flow: acquire -> clean -> quality agent -> gold -> model -> index. The quality agent logs every FIX / FLAG / FAIL decision with a reason; it never invents missing observations. `make all` rebuilds the data layer, refits the model, rebuilds the index, runs tests, regenerates this PDF and exports prefect_runs.csv. Pipeline history contains only real runs with true timestamps; the missed Tue-Wed window is disclosed, not backfilled.

Honesty controls

Quality status: WARN (13 rules; failures are surfaced, warnings are kept). Gold-table contract is enforced by the loader. Post-hoc verification rejects any draft answer containing an unsupported number or unknown citation. The offline mode is extractive, so hallucination is impossible by construction. The mock fixture is labelled MOCK and never evaluated.

Pakistan pass-through (small sample, descriptive)

39 overlapping weeks; correlation 0.6928; slope 0.5821 PKR per USD/bbl. Not a causal claim; domestic prices are administered.

Limitations

Short Pakistani weekly history with genuine publication gaps; administered domestic pricing; the primary model targets global Brent volatility rather than the domestic retail level; lexical retrieval fallback without a Mistral key; frontend freshness is bounded by the latest collected PBS week.